8.3 Test Monitoring and Control

3

Test Monitoring and Control

- The purpose of test monitoring is to gather information and provide feedback and visibility about test activities.
- Test control describes any guiding or corrective actions taken as a result of information and metrics gathered and (possibly) reported



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4

Test Control Activities Example

- Re-prioritizing tests when an identified risk occurs
- Changing the test schedule due to availability or unavailability of a test environment or other resources
- Re-evaluating whether a test item meets an entry or exit criterion due to rework



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5

Test Control (1)

- If there are delays compared to the project plan or test plan, adequate corrective measures are to be taken. For example:
  - If (newly) recognised risks appear, change the prioritisation of tests (e.g. delayed delivery of software parts)
  - Adjustment of the time planning, if the availability of the test environment causes delays
  - Corrected defects are to be re-tested by a developer before the software is taken further
- If needed, request and deploy additional test resources (staff, work space, tools) in order to catch up with the test plan in remaining iterations

6

Test Control (2)

- If no additional resources are available, the test plan must be adjusted
  - Low priority test cases may be cancelled
  - Test cases which are designed in different variants may be executed only for one variant (e.g. tests are only executed in one operating system instead of different ones as originally planned)
  - As a result of these adjustments, some interesting test cases may not be executed, but the saved resources will enable at least the high priority test cases to be executed

Test Control (3)

- Depending on how serious the detected defects and issues are, the test duration can be extended to allow for additional test iterations which may be needed After each correction phase, the changed software has to be re-tested
- This can lead to a delay of the deployment or the delivery of the software product
- It is important, that the test manager documents and communicates all changes to the plan Changes in the original test plan mean generally an increase in the release risk
- It is the responsibility of the test manager to present the risk to the project manager openly, clearly and in a timely manner.

Test Control (Summary)

- Test control describes any guiding or corrective action taken by the test manager as a result of metrics or information gathered and reported
- Measures may address any test activity and may affect any other software lifecycle activity or task.
- Examples of measures for test control are:
  - Re-prioritisation of tests if new or changed risks occur
  - Changes of the test schedule based on the availability of the test environment
  - Setting of an entry criteria that requires defect corrections to be re-tested before the software is given to the next test level

9

Common Test Metrics

- Percentage of planned work done in (test case preparation/implementation-Test environment preparation)
- Test case execution (e.g., number of test cases run/not run, test cases passed/failed)
- Defect information
- Test coverage of requirements, user stories, acceptance criteria, risks, or code
- Task completion, resource allocation and usage, and effort
- Cost of testing

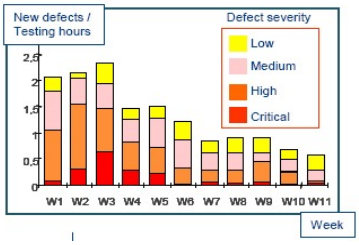


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10

Test Progress Monitoring

- Monitoring of progress and test objectives and the corresponding test exit criteria which are defined in the test plan based on objective test metrics.
- Only such metrics are to be used, which can be regulary, reliably and simply measured. Their informative value must be clear and give no room for incorrect interpretation.
- Frequently used test metrics are among others
  - Defect-based metrics, e.g.
    - defect detection percentage
    - found and resolved defects
    - degree of defect exposure
    - results of re-tests

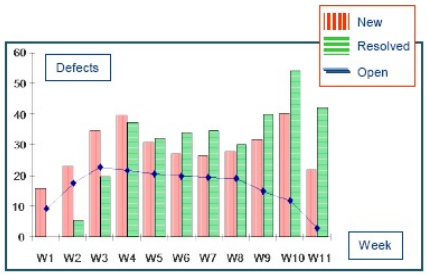


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11

Test Progress Monitoring

- Test case-based metrics, e.g.
  - Relation between already specified test cases and the total amount of test cases planned
  - Quantity of executed/not executed test cases
  - Relation of efforts required to establish the test harness and the total testing efforts
- Test object-based metrics e.g.
  - Test coverage of requirements, risks or code
- Cost based-metrics e.g.
  - Comparision of the costs for finding the next defect and the costs of the next test execution
- Subjective tester's trust into the product (not traceable!) – needed in case of exploratory testing



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12

Test Reporting

- The purpose of test reporting is to summarize and communicate test activity information, both during and at the end of a test activity
- The test report prepared during a test activity may be referred to as a test progress report, while a test report prepared at the end of a test activity may be referred to as a test summary report
- typical test progress reports may also include:
  - The status of the test activities and progress against the test plan
  - Factors impeding progress
  - Testing planned for the next reporting period
  - The quality of the test object



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13

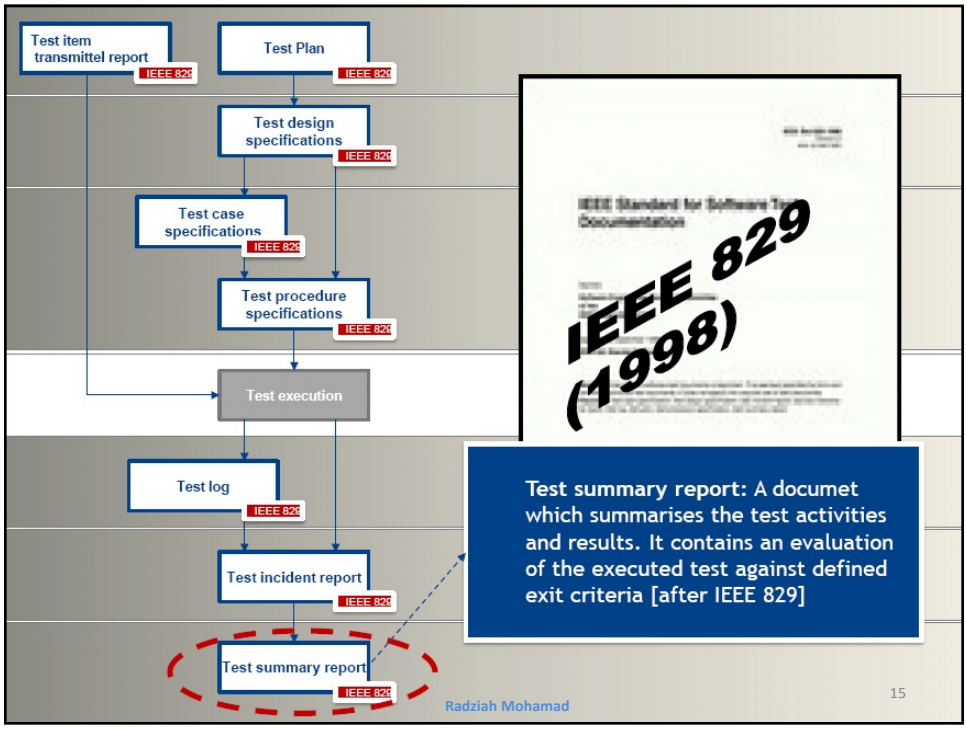
Test Reporting

- When exit criteria are reached, the test manager issues the test summary report.
- This report provides a summary of the testing performed, based on the latest test progress report and any other relevant information



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14


Test item transmittel report
Test Plan
Test design specifications
Test case specifications
Test procedure specifications
Test execution
Test log
Test incident report
Test summary report
IEEE 829 (1998)
Test summary report: A documet which summarises the test activities and results. It contains an evaluation of the executed test against defined exit criteria [after IEEE 829]
Radziah Mohamad
15

15